

Link to Working System

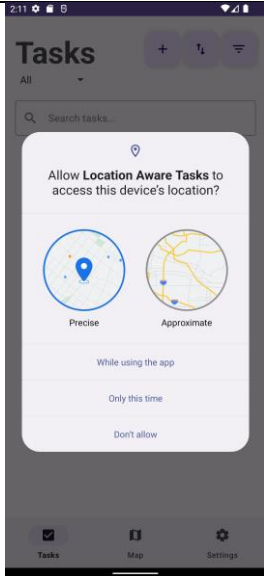
The link to the GitLab repository is <https://code.vt.edu/schamilton/location-aware-tasks>.

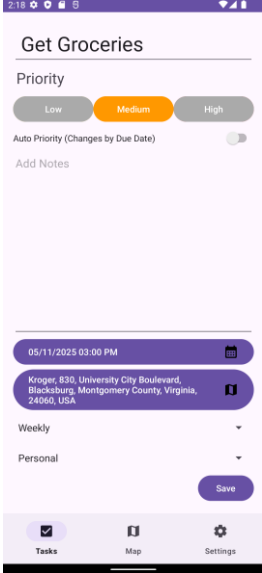
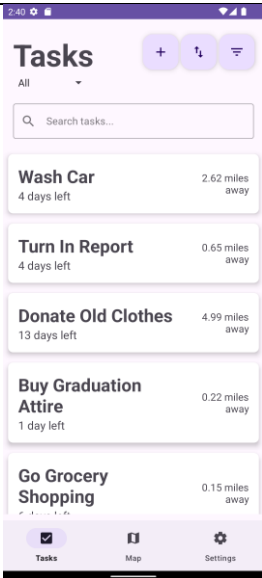
The link to the demo recording on repository is <https://code.vt.edu/schamilton/location-aware-tasks/-/wikis/Full-App-Demo>.

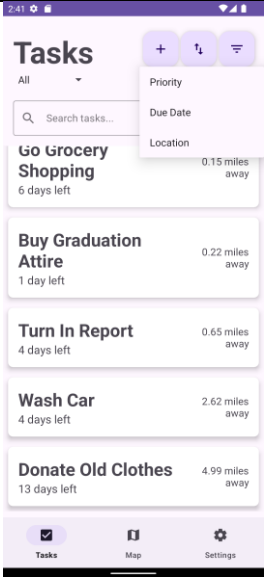
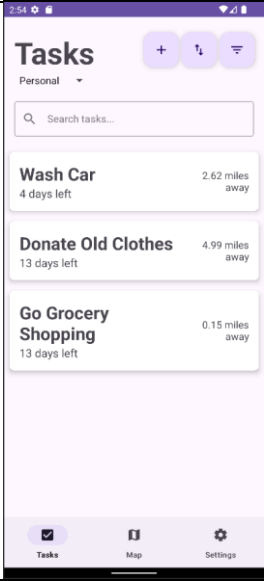
Brief Product Description.

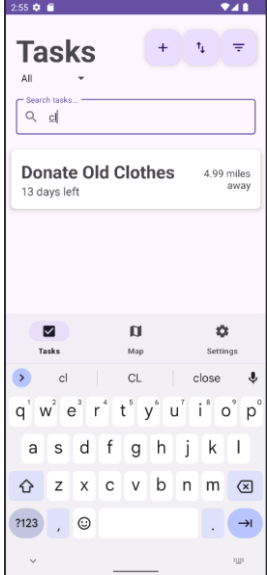
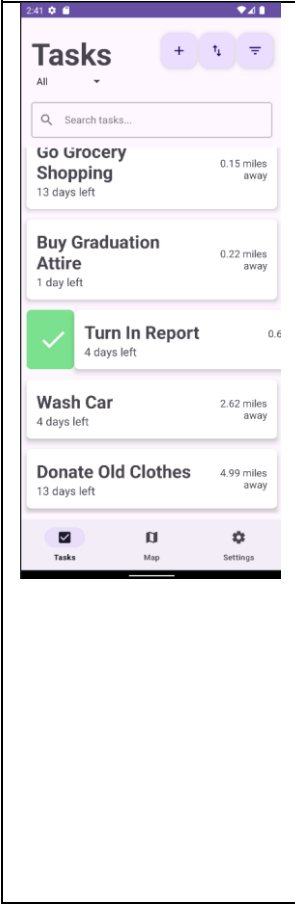
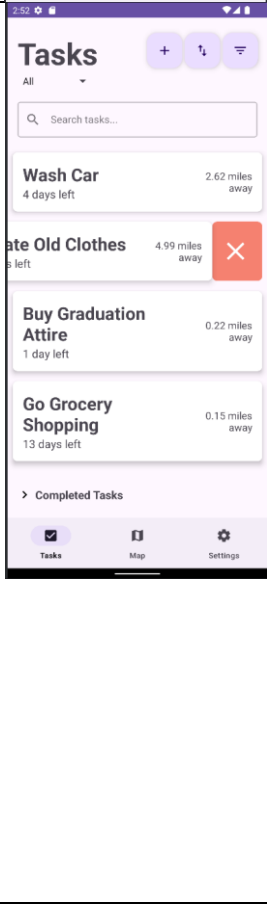
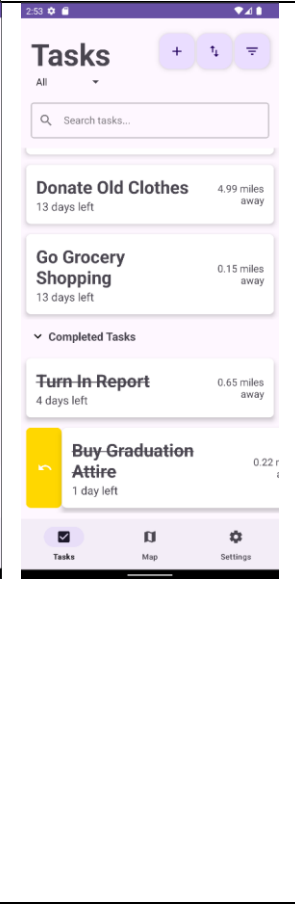
Our product is a location-aware task management app that helps users stay organized and productive. Users can create tasks with notes, due dates, priority levels, and assign each task a specific location. An interactive map view lets users visualize where their tasks are and plan their day accordingly. With intuitive swipe actions to complete or delete tasks, built-in sorting, and a search feature, managing to-do lists becomes seamless. Our app also supports importing/exporting tasks for easy sharing and notifies users when they have an upcoming due date.

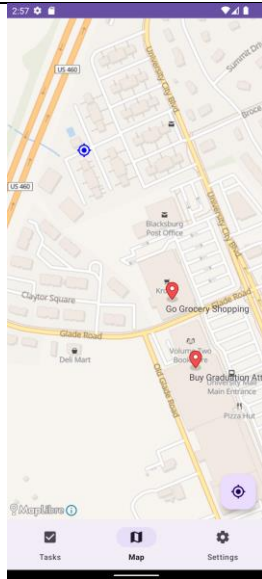
Product Functionalities.

App Screenshot	App Description
	Users are prompted to enable location services for the app upon running it for the first time.

	During the create a task view, users can: - Name the task - Set priority for the task, or auto-priority - Add notes for the task - Set due date and time - Set location for task - Set task repetition time - Set task type
	On the task list view, users can perform a variety of options: - Plus button: create a new task - Switch button: reverse the order of the list - Inverted triangle button: sort tasks in list - All dropdown button: filter tasks by type - Search bar: search for specific tasks - Click on existing task: click on existing task to edit in create a task view

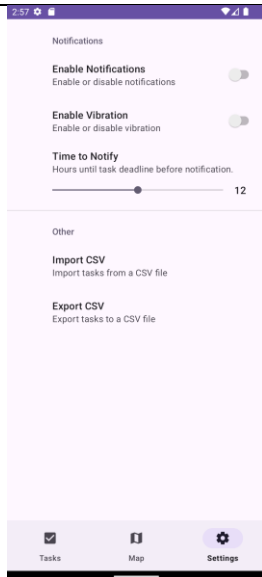
	This is a demonstration of the sort button. In this photo the user sorted by location.
	This is a demonstration of the filtering tasks by type. In this photo the user filtered by Personal tasks.

			This is a demonstration of the search bar. In this photo the user started typing the letters “cl” and the view dynamically updated to show tasks whose names contain that combination of letters.
			On the task list view, users can also perform three actions on each task: - Swipe right to set as complete: swiping right from main task list sets task as complete and removes from view - Swipe left to delete: swiping left deletes task and removes from view - Swipe right to set as incomplete: swiping right from completed task list sets task as incomplete and adds to main list



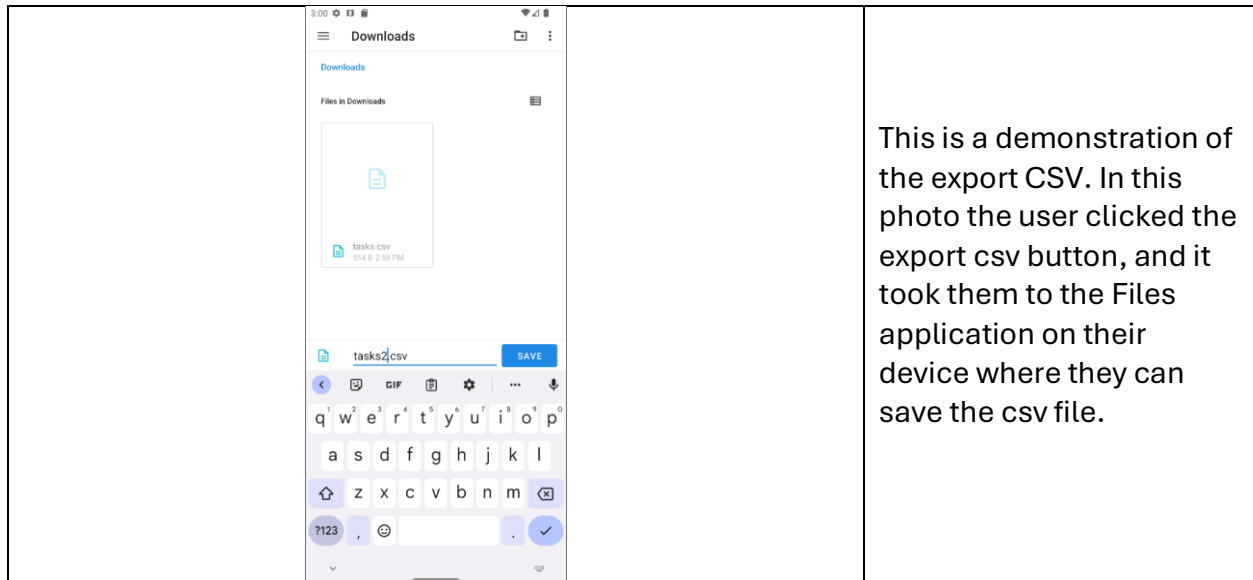
On the map view, users can perform a variety of options:

- Click and drag: moves the map around to see more map area
- Click and hold: holding spot on location pulls up create a task view with preset location.
- Click on existing task: click on existing task to edit in create a task view



On the settings view, users can perform a variety of options:

- Enable notifications: click to enable app notifications
- Enable Vibration: click to enable app vibration
- Time to Notify: change slider to change notification timing
- Export/Import CSV: export/import tasks as a csv file.



Design.

Unique Features

Our application is designed to help people keep track of their pending tasks in a clean and organized manner. Our product is unique because it allows users to set locations for their tasks, making it easier for users to complete their pending tasks while going about their day-to-day activities. Our design stands apart from other task organizing apps because we prioritize location as one of the main organizing details. On our main task list view, users are able to see their current distance from each of their task locations as well as the due date. This intuitive design makes it very easy for users to keep up with their tasks and complete them in an efficient manner. Besides that, there are also many other features on our app that allow users to stay as organized as possible while also keeping our app easy and simple to use.

App Features

On the task list view, users can sort the current tasks based on a variety of options, such as due date, location, or priority, as well as reverse the order of the current list. Users also have the option to search for specific tasks using the search bar at the top, which dynamically updates the list view as the user types. Aside from that, users can also filter out their tasks based on what list it belongs to, such as default, work, or personal. As for the actual task list, our app makes it very easy for users to complete or delete tasks. If users swipe right on a task, it sets it to complete and sends it to the expandable container

at the bottom of the screen. Users can easily undo this option by going into the container at the bottom and swiping right on the task again. If users swipe left on a task, it deletes it, and a toast pops up temporarily at the bottom of the screen giving user the option to undo the action if needed. Tasks are also able to be edited by pressing on the task once. Finally, users are also able to create new tasks by pressing the add task button which is also on this screen.

On the map view, users can move around the map to see all their pending tasks and their locations. Users are also able to hard-click on a spot on the map and create a new task with that specific location. Besides that, users can also edit current tasks by pressing on the task from this view. We chose to include this view along with the task list view because some users prefer to see a visual layout of their tasks. This view also allows users to plan out the route to complete their tasks.

Our task creation page allows users to create new tasks with a variety of features. These include name, notes, priority level, due dates, repetitions, list, and location. After a user enters all the relevant information, they are able to press save and the task instantly shows up on both the task list view and the map view. Our tasks are stored locally using the Room database.

Finally, the last page in our design is the settings page. On this page, users can change their notification settings for tasks based on time from due date or distance from task location. Aside from that, users are also able to import and export their tasks as csv files.

Location Integration

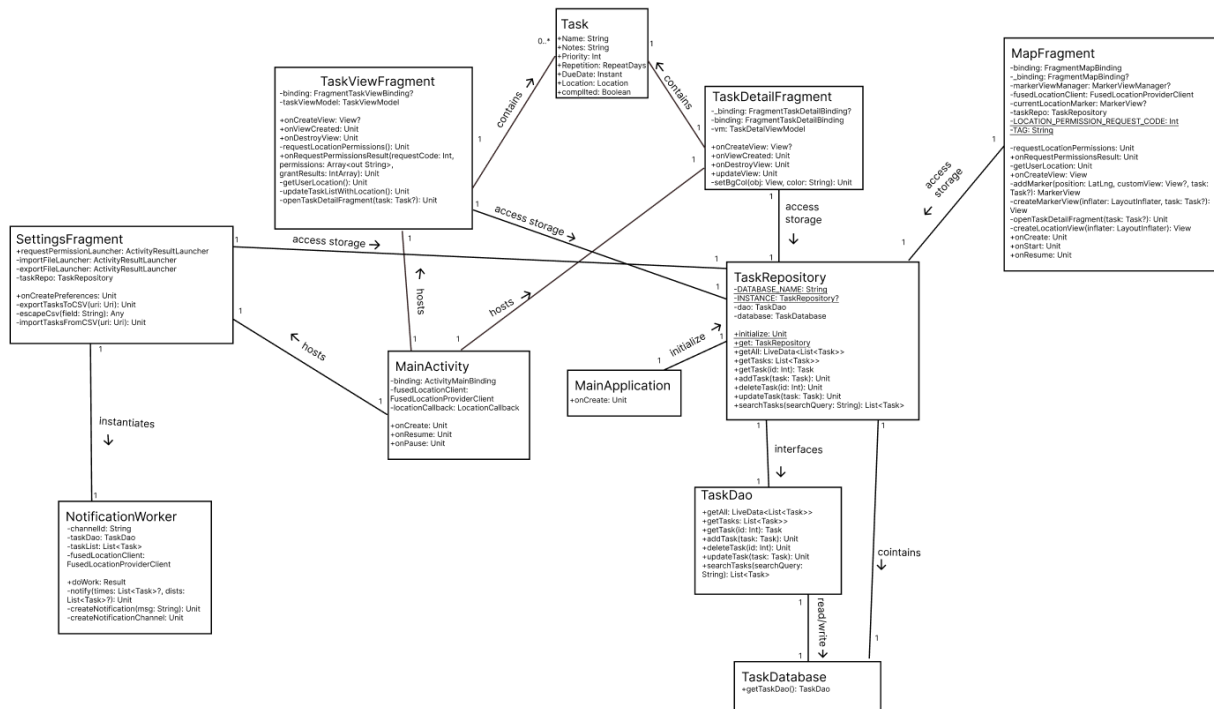
To make our app location-aware, we used the LocationIQ API for geocoding, which is the process of turning addresses into latitude/longitude, as well as reverse-geocoding, which does the opposite. When users needed to select a location for their task, they were able to search for an address using a text box. As they typed, we used LocationIQ's API to fetch live suggestions. Then once a location was selected, it's latitude and longitude were stored as a structured Location object in the Task database.

Architecture and Frameworks

Our application was built in Android Studio using Android's native development framework with Kotlin, ensuring smooth performance and successful integration with Android device features such as location services and notifications. We followed the MVVM (Model-View-ViewModel) architecture pattern, which allowed us to cleanly separate the UI from the

application logic, simplify data binding between the UI and logic layers, and improve testability. Finally, we handled data flow in our app across the different components though LiveData and ViewModel observers.

Design Class Diagram



Ethical Implications.

Our app handles sensitive user data such as physical location and task information, which raises important ethical considerations around privacy, data security, and user consent. To mitigate these concerns, we designed the app to store all user data locally on the device. This ensures that no personal task or location information is transmitted to external servers, giving users full control over their data. For data security, we use Room, Android's official local database library, which provides a reliable and efficient way to store structured data while minimizing the risk of data leaks. Additionally, we address user consent by explicitly requesting location permissions upon first launch. This ensures that users are fully informed and able to choose whether or not to grant location access. Finally, users have the option at any time to delete the application from the device, which effectively deletes and stored data permanently.

Retrospection.

The process of building our application was very successful for the most part. We were able to accomplish almost all the user stories we had set for ourselves and had very few setbacks. One difficulty we faced during the initial app building process was the location integration. It was a bit difficult to find a free location related API that would be able to provide all the functionalities we were looking for. We initially had wanted to go with the Google Maps API since we knew it would look very clean and function smoothly, but we were not able to find a free version. For this reason, we eventually ended up using the LocationIQ API, which gave us all the functionalities we needed. Our project building process was mostly successful because of our clear communication within our team and our organization. Using Gitlab as our project management tool made it very easy for us to keep track of our own tasks each sprint and view our team's progress as a whole. Finally, our consistency with our weekly meetings made it easy for our team to check up on each other and stay on track with our deadlines.

Recommendations For the Future.

In the future, we could improve this product by allowing users to create accounts. Doing this would open up quite a few possibilities for this app, such as collaboration on certain tasks between users or storing user data and their task information on a secure remote database. Another feature we could add to this app would be the ability to suggest optimal paths between certain tasks, which would make the app more helpful for users who need help planning errands. Finally, the last feature we would add in the future would be more accessibility support, such as increased contrast.